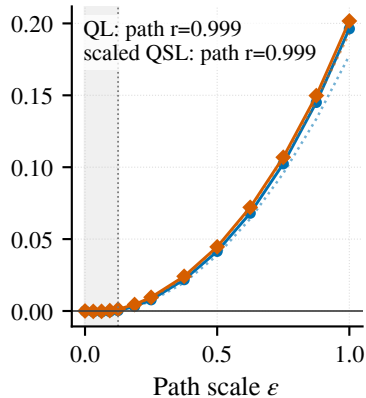


Held-out  $\Delta\text{NLL}$  (nats/token)

(a)

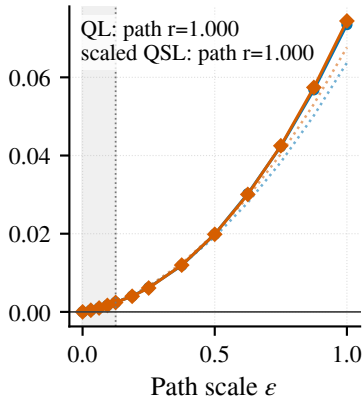
Pythia-70M

● Q+L      — — — fit:  $\varepsilon \leq 0.125$   
◆ strict scaled QSL      ····· fit extrapolation



(b)

OPT-125M



(c)

Qwen3-0.6B

